u

$[F]_{n_x, n_y} = e^{-j2\pi(n_x-1)(n_y-1)/N}$

$\downarrow$ 2D Fourier

$FF^H = NI$. $F^{-1} = \frac{1}{N}F^H$. $F = F^T$. $\bar{F} = F^H$

$\hat{u} = FuF^H$ $\longrightarrow$ 中心化 $\hat{u}_c = J\hat{u}J^H$

频域称 k-space

$\begin{bmatrix} & I_{N/2} \\ I_{N/2} & \end{bmatrix}$

$\longrightarrow$ mask 掩膜 $M \odot \hat{u}_c$

即 Subsample，非零观测 $V = S\hat{u}_c S^H + E$

$[0, I_M, 0]_{M\times N}$ $M = N f_{us}$

降采率

$V = SJF \cdot u \cdot (SJF)^H + E = \Phi u \Phi^H + E$

$\Phi = SJF$

$v = (\bar{\Phi} \otimes \Phi)u + e = A_{M^2\times N^2} u + e$